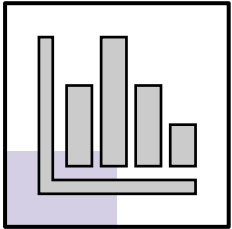


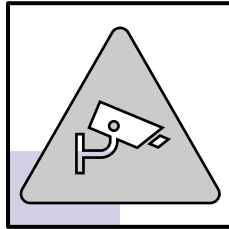
Privacy in Sensor-Driven AIoT Systems

Why Privacy Matters : AIoT sensors capture highly sensitive personal and behavioral data.



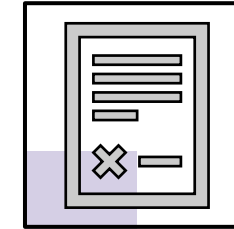
Sensitive Data Collected

- Location & movement patterns
- Health & biometric metrics
- Video & facial imagery
- Behavioural & usage patterns



Privacy-by-Design Approaches

- Data minimization (collect only what is necessary)
- Anonymization & pseudonymization
- Edge processing to keep data local



Transparency & Consent

- Inform users about data collection & usage
- Obtain consent & ensure lawful processing

Privacy-aware design builds trust and adoption

Key Privacy Challenges in Sensor-Driven AIoT



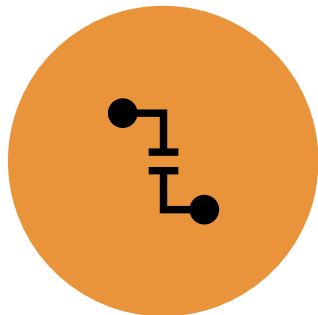
Excessive and Continuous Collection :

Sensors (e.g., accelerometers, cameras, and microphones) often gather far more information than users realize, creating massive digital footprints.



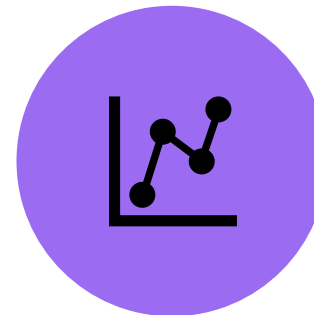
Lack of Awareness and

Consent: Pervasive sensing often occurs without explicit user permission, and consent mechanisms are frequently buried in complex, lengthy policies.



Data Inference and Profiling:

AI models can analyze seemingly harmless sensor signals (like motion data) to infer highly sensitive personal information without user participation.



Identification and

Localization: Aggregated data from various sensors can be used to re-identify anonymous users and track their past and present physical locations precisely.

Privacy risks extend beyond the data users knowingly share

Privacy-Preserving Techniques

Why Privacy Matters : AIoT sensors capture highly sensitive personal and behavioral data.

Differential Privacy

- Adds statistical noise to data outputs
- Protects individual identity while preserving trends

Federated Learning

- Data stays on local devices
- Only model updates are shared
- Reduces exposure of sensitive information

Anonymization & Obfuscation

- k-anonymity masks identifying attributes
- Signal transformation hides sensitive patterns
- Maintains AI usefulness while protecting privacy

Cryptographic Protection

- Homomorphic Encryption enables computation on encrypted data
- Lightweight cryptography secures low-power devices

Edge & Fog Processing

- Process sensitive data closer to the source
- Reduces exposure & transmission risks

Enable intelligent insights while preserving user privacy

Regulatory & Ethical Safeguards in AIoT

1

Privacy by Design

- Integrate privacy protection from system inception
- Build security & privacy into architecture

2

Regulatory Compliance

- Align with global standards & sector regulations
- Ensure lawful data collection & processing

3

Transparent Data Practices

- Clear data usage & retention policies
- Inform users how data is collected & used

4

Access Control & User Rights

- Multi-Factor Authentication & role-based access
- Tools for users to view, manage & delete data

Trustworthy AIoT systems balance innovation with responsibility